

Surname	Centre Number	Candidate Number
First name(s)		0



GCSE

3445UC0-1



FRIDAY, 10 MAY 2024 – MORNING

APPLIED SCIENCE (Double Award)
UNIT 3: Food, Materials and Processes
HIGHER TIER

1 hour 30 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	9	
2.	10	
3.	7	
4.	11	
5.	7	
6.	13	
7.	8	
8.	10	
Total	75	

ADDITIONAL MATERIALS

In addition to this examination paper, you will require a calculator, pencil and a ruler.

INSTRUCTIONS TO CANDIDATES

- Use black ink or black ball-point pen. Do not use gel pen or correction fluid.
- You may use a pencil for graphs and diagrams only.
- Write your name, centre number and candidate number in the spaces at the top of this page.
- Answer **all** questions.
- Write your answers in the spaces provided in this booklet.

INFORMATION FOR CANDIDATES

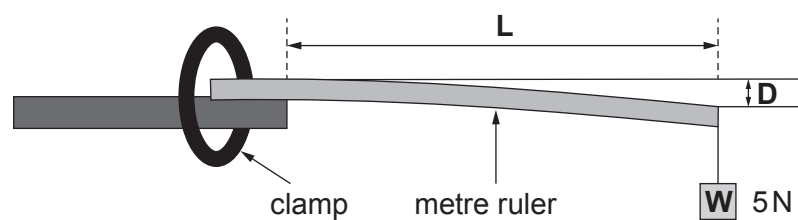
- The number of marks is given in brackets at the end of each question or part-question.
- Question **7(a)** is a quality of extended response (QER) question where your writing skills will be assessed.
- You are reminded to show all your workings. Credit is given for correct workings even when the final answer given is incorrect.
- The Periodic Table is printed on the back cover of the examination paper.



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Answer **all** questions.

1. Students are investigating how the stiffness of a wooden metre ruler depends on its length. They use the apparatus shown below. The weight, **W**, on the end of the ruler remains constant at 5 N.



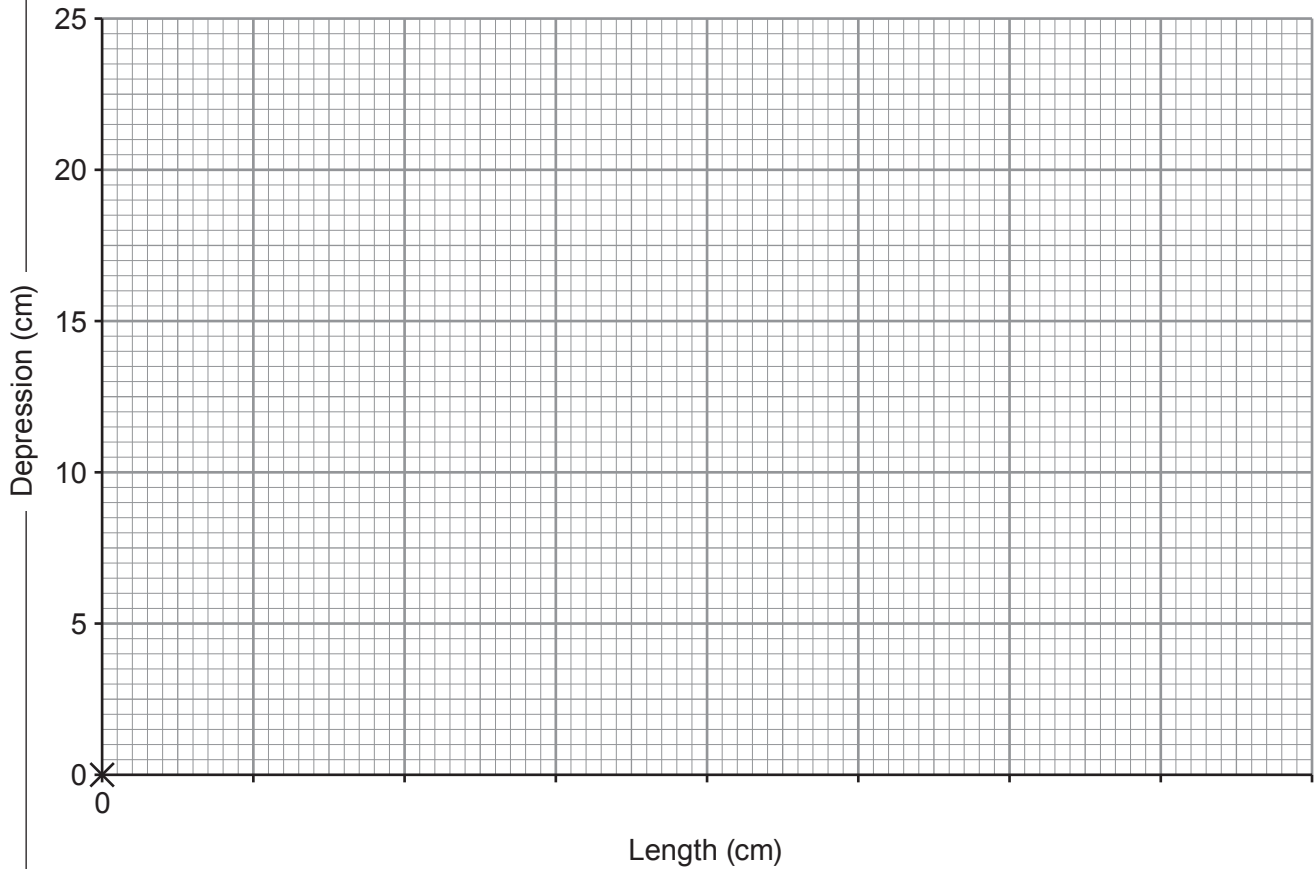
The length, **L**, is varied. For each length, **L**, the depression, **D**, of the end of the ruler is measured.

Their results are shown in the table below.

Length L (cm)	Depression D (cm)
0	0.0
30	1.0
40	3.0
50	6.0
60	10.0
70	15.0
80	22.0



- (a) Plot the data on the grid below and draw a smooth curve. The point (0,0) has already been plotted for you. [4]



- (b) It was expected that depression, **D**, would be directly proportional to length, **L**. Explain whether the graph shows a proportional relationship. [2]

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- (c) Use information in the table and the equation:

$$\text{stiffness} = \frac{\text{weight}}{\text{depression}}$$

to calculate the stiffness of the ruler at a length of 50 cm. [3]

stiffness = N/cm



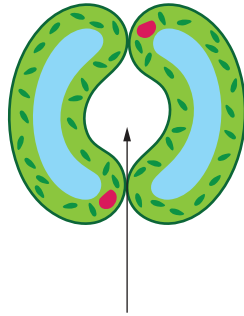
2. Photosynthesis is the process by which plants make food from water and carbon dioxide.

- (a) Stomata are an important part of the leaf. In many plants, stomata remain open during the day and closed at night.

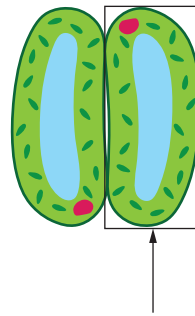
- (i) Label the diagram below.

[2]

Open by day



Closed by night



- (ii) State **two** functions of stomata.

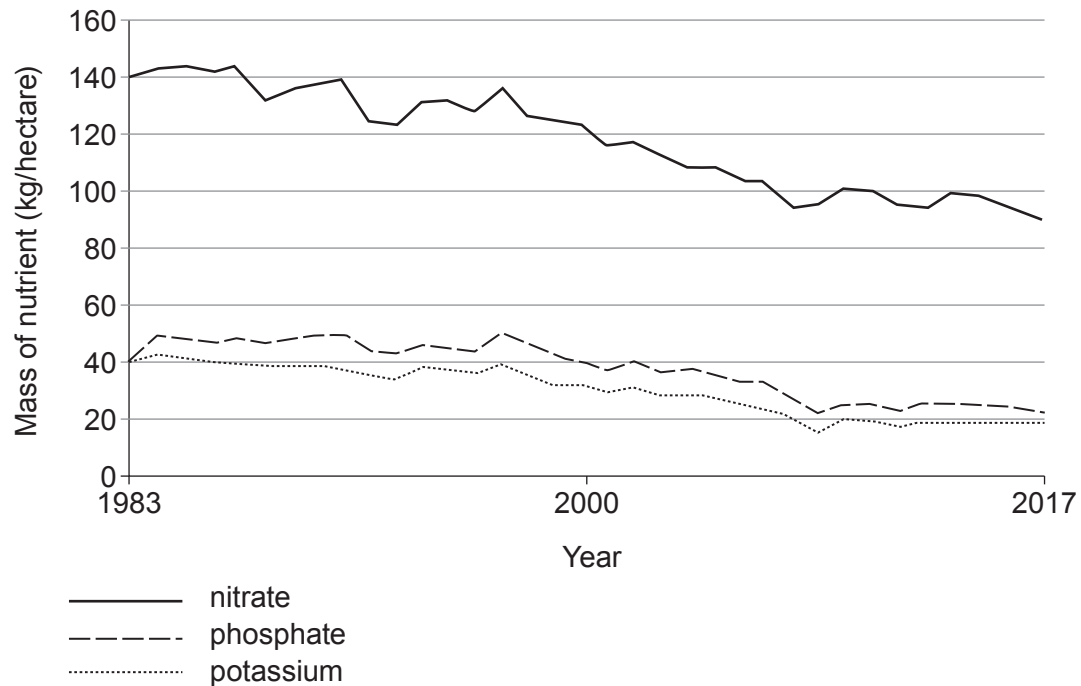
[2]

1.
2.



- (b) Healthy plants require a supply of nutrients. Many farmers use artificial fertilisers to supply these nutrients to the soil.

The graph shows the mass of different nutrients applied to agricultural land between 1983 and 2017.



Use the data from the graph to answer the following questions.

- (i) It is claimed that the use of nitrate fertilisers halved between 1983 and 2017. Use the graph to explain whether you agree.

[3]

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- (ii) The total land area of Wales is approximately 2 100 000 hectares.
60 % of this land is used for farming.

I. Calculate the area of land used for farming in Wales.

[1]

area = hectares

II. Calculate the total mass of phosphates used in farming in Wales in 1983. [2]

mass = kg

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3. Qualitative analysis is used to identify the ions present in a compound.

- (a) The table shows the observations from chemical tests carried out on solutions of two compounds.

Sample	Test	Observation	Name of compound tested
1	add dilute acid	colourless gas evolved which turns limewater milky	
	flame test	lilac	
2	flame test	lilac	
	add nitric acid then silver nitrate	white precipitate	

Complete the table.

[3]

- (b) Another compound was identified as calcium sulfate (CaSO_4).
Use the information below and the equation:

$$\text{number of moles} = \frac{\text{mass}}{M_r}$$

to calculate the number of moles of calcium sulfate present in 34 g of powder.

[4]

Element	A_r
Calcium	40
Sulfur	32
Oxygen	16

number of moles =



4. Yeast is used in fermentation to make the ethanol in wine.

(a) Write down the word equation for fermentation. [1]

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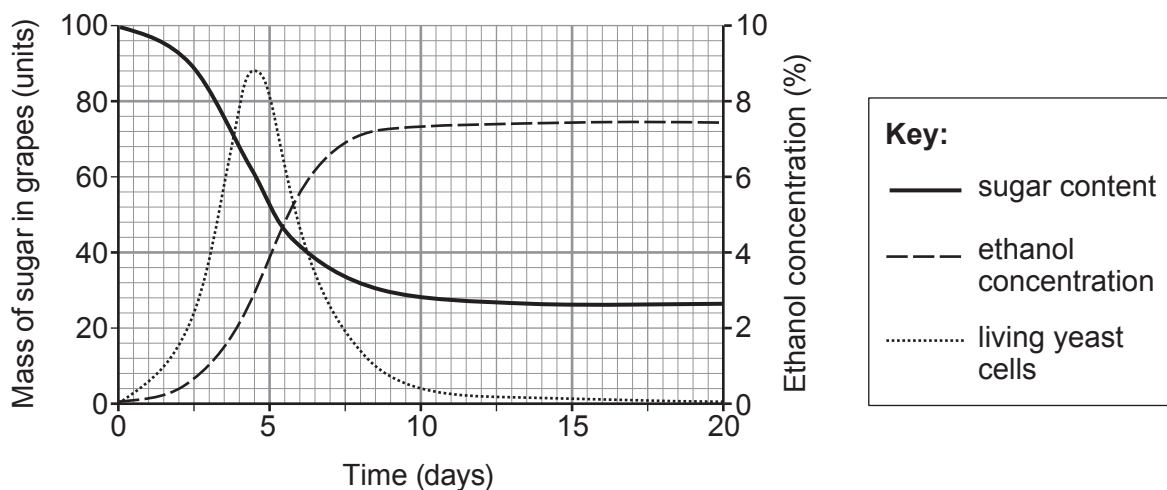
(b) Genetically modified yeast can be used in wine making.

(i) State the meaning of the term 'genetically modified'. [1]

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(ii) One type of genetically modified yeast was used to make wine. The graph below shows how sugar content (solid black line), ethanol concentration (dashed line) and number of living yeast cells (dotted line) change with time.



A home brewer thinks that when the number of living yeast cells is at a maximum, the sugar content will be at its minimum and the ethanol concentration would be at its maximum. Use data from the graph to explain whether you agree. [3]

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- (c) Many countries have banned the use of genetically modified foods.

Suggest **two** reasons for their concern.

[2]

1.

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2.

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- (d) The home brewer also uses this genetically modified yeast to produce beer. Complete the steps in the fermentation process. Three steps have been given. [4]

Step 1: Barley is mixed with water. Starch in the barley is converted to glucose.

Step 2: Hops are added for flavour.

Step 3:

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Step 4:

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Step 5:

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Step 6:

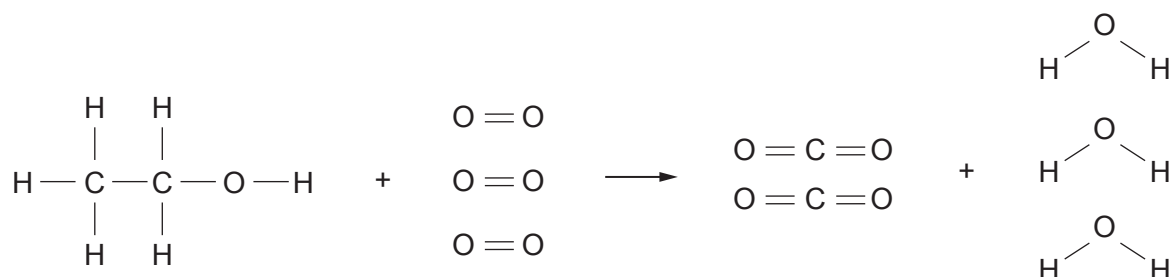
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Step 7: The beer may be pasteurised, then stored in bottles or casks.



5. Ethanol is widely used in biofuels. When ethanol is burned in a car engine, carbon dioxide and water are formed.

The displayed formula equation for this reaction is shown below.



Some of the bond energies for this reaction are shown in the table below.

Bond	Bond energy (kJ/mol)
C–C	347
C–H	413
C=O	732
C–O	336
O–H	464

Use the information above to answer the following questions.

- (a) The energy needed to break all the bonds in the reactants is 4706 kJ. Calculate the energy needed to break an O=O bond. [4]

energy = kJ



(b) The energy released to form new bonds in the products is 5712 kJ/mol.

Explain why a car engine using ethanol as a fuel needs a cooling system.

[3]

Examiner
only

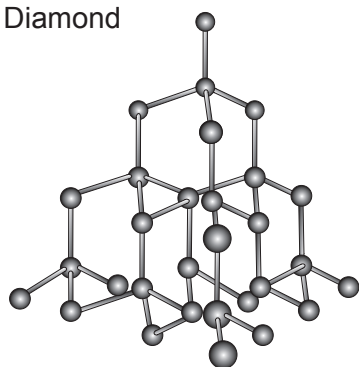
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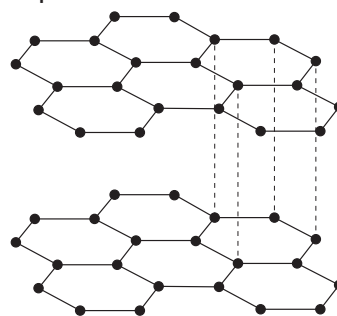
6. Carbon exists as a number of different allotropes. These allotropes have some important commercial uses. The uses of these allotropes depends upon the structure.

(a) Two allotropes of carbon are diamond and graphite.

Diamond



Graphite



Use the diagrams and your knowledge to answer the following questions.

- (i) **Describe** the bonding between a carbon atom and its nearest neighbours in diamond.
Include a dot and cross diagram of one carbon atom and its nearest neighbours in your answer. [3]

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- (ii) State the difference in the bonding between carbon atoms in diamond and graphite. [1]

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Examiner
only

(iii) Explain why graphite is a good conductor of electricity but diamond is a good insulator of electricity. [3]

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(iv) Explain, in terms of structure and bonding, why graphite is soft and slippery but diamond is hard. [3]

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(b) Graphene is a single layer of graphite. It can be rolled into the shape of a cylinder which is called a carbon nanotube.

The table gives some data about carbon nanotubes, steel and aluminium.

Material	Strength (MPa)	Density (g/cm ³)	Maximum length of power line before it breaks (km)	Electrical conductivity (× 10 ⁶ units)
carbon nanotube	62 000	1.3	4700	>> 1000
steel	500	8.0	6.4	7
aluminium	80	2.7	1.0	40

Electricity power lines are currently made from a combination of steel and aluminium. Explain why carbon nanotubes would be better. [3]

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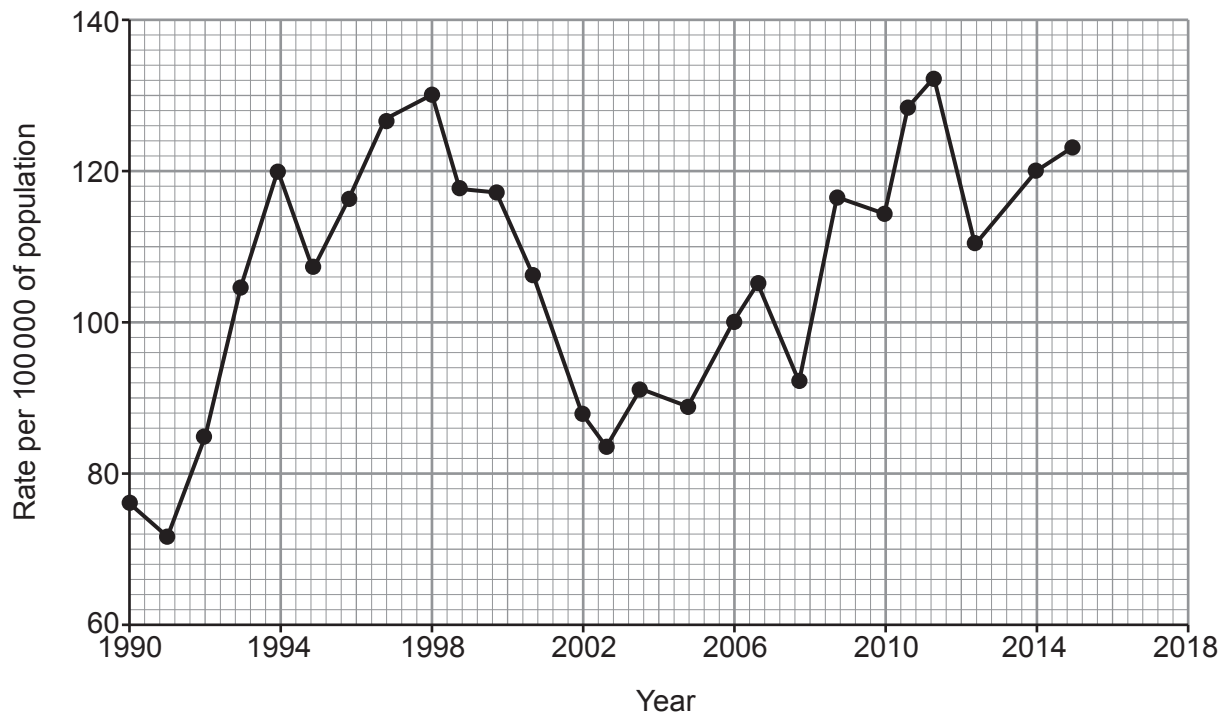
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8. The graph below shows the confirmed cases of *Campylobacter* food poisoning per 100 000 of the population in Wales from 1990–2015.



In Wales, for every confirmed case of *Campylobacter* poisoning there are another two unconfirmed cases. The population of Wales is 3 million.

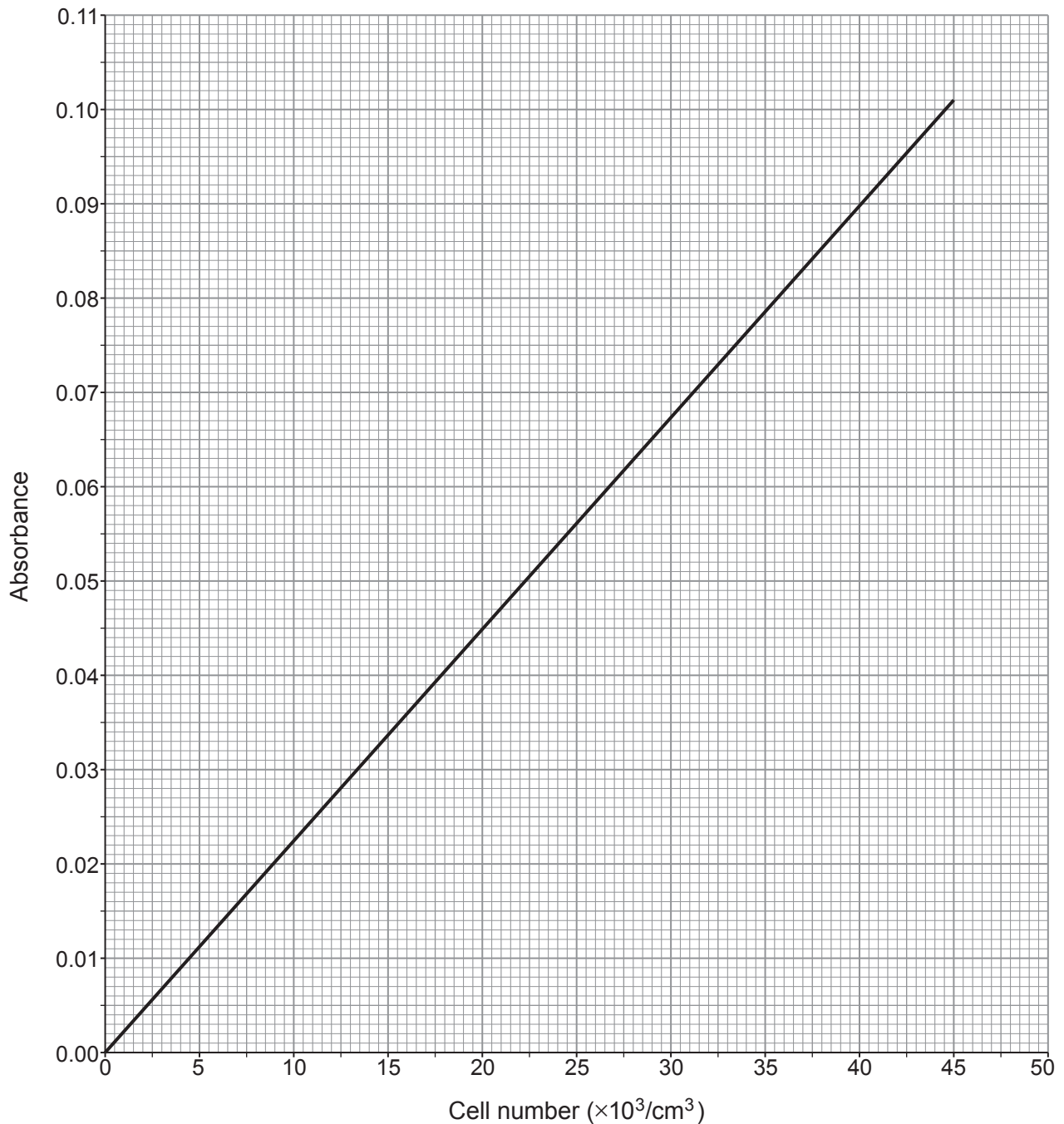
- (a) Calculate the total number of *Campylobacter* cases in Wales in 1998.

[3]

total number of cases =



- (b) Turbidity can be used to calculate the number of bacteria present in a sample. Alun used turbidity to study the growth rate of *Campylobacter* in water. He used the following standard graph.



(i) Alun analysed a water sample. The sample gave an absorbance of 0.01.

I. Use the graph to find the cell number per cm^3 at an absorbance of 0.01. [1]

cell number = per cm^3

II. *Campylobacter* reproduce every 20 mins The sample was left to stand for 1 hour. Alun thought that the absorbance would be 0.03 after 1 hour. Use your knowledge and data in the graph to explain whether he is correct. [3]

Space for calculations.

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(ii) The threshold value for making a person sick is 120 000 bacteria. In (b)(i)I. Alun's sample was 7 cm^3 . Use your answer above to determine whether this sample of water would make a person sick after standing for the hour. [3]

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END OF PAPER



Group

02

1 H Hydrogen 1																								
7	Li	9								11	B	12	C	14	N	16	O	19	F	20	Ne			
	Lithium	Beryllium								Boron	Carbon	Nitrogen	Oxygen	Fluorine	Neon									
3		4								5	6	7	8	9	10									
23	Na	24								27	Al	28	Si	31	P	32	S	35.5	Cl	40	Ar			
	Sodium	Magnesium								Aluminium	Silicon	Phosphorus	Sulfur	Chlorine	Argon									
11		12								13	14	15	16	17	18									
39	K	40	45	48	51	52	55	56	59	59	63.5	65	70	73	75	79	80	84						
	Potassium	Calcium	Scandium	Titanium	Vanadium	Chromium	Manganese	Iron	Cobalt	Nickel	Copper	Zinc	Gallium	Germanium	Arsenic	Selenium	Bromine	Krypton						
19		20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36						
86	Rb	88	89	91	93	96	99	101	103	106	108	112	115	119	122	128	127	131						
	Rubidium	Strontium	Yttrium	Zirconium	Niobium	Molybdenum	Technetium	Ruthenium	Rhodium	Palladium	Silver	Cadmium	Indium	Tin	Antimony	Tellurium	Iodine	Xenon						
37		38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54						
133	Cs	137	139	179	181	184	186	190	192	195	197	201	204	207	209	210	210	222						
	Caesium	Barium	Lanthanum	Hafnium	Tantalum	Tungsten	Rhenium	Osmium	Iridium	Platinum	Gold	Mercury	Thallium	Lead	Bismuth	Polonium	Astatine	Radon						
55		56	57	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86						
223	Fr	226	227																227	226	223			
	Francium	Radium	Actinium																Actinium	Radium	Francium			
87		88	89																89	88	87			
Key																								

relative atomic mass

atomic number